

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

PRACTICAL - 1

AIM :-

To implement Breadth First Search (BFS) algorithm for finding a route from Home to College using the available road network. .

Code :-

1. Practical_1 Code.py

```
# T101 RAJDEEP M PARAB
# Breadth First Search (BFS)
# Problem: Find a route from Home to MVLU College
import queue as Q
# ROAD NETWORK
dict_gn = dict(
    Home=dict(
        Matharpada_Rd=110
    ),
    Matharpada_Rd=dict(
        Amit_Naik_Marg=87
    ),
    Amit_Naik_Marg=dict(
        Caesar_Cross_Rd_No_1=110
    ),
    Caesar_Cross_Rd_No_1=dict(
        Caesar_Rd=58,
        Caesar_Rd_Alternative=180
    ),
    # Route 1
    Caesar_Rd=dict(
        Swami_Vivekanand_Rd=240
    ),
    # Route 2
    Caesar_Rd_Alternative=dict(
        Swami_Vivekanand_Rd=650
    ),
    Swami_Vivekanand_Rd=dict(
        Model_Co_op_Bank=400
    ),
    Model_Co_op_Bank=dict(
        Andheri_Subway_Rd=400
    ),
    Andheri_Subway_Rd=dict(
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

```
Dr_S_Radha_Krishnan_Rd=500
),
Dr_S_Radha_Krishnan_Rd=dict(
    MVLU_College=230
),
MVLU_College=dict()
)
```

TWO AVAILABLE ROUTES

```
route1 = [
    "Home",
    "Matharpada Rd",
    "Amit Naik Marg",
    "Caesar Cross Rd No. 1",
    "Caesar Rd",
    "Swami Vivekanand Rd",
    "Model Co-op Bank",
    "Andheri Subway Rd",
    "Dr. S. Radha Krishnan Rd",
    "MVLU College"
]
route2 = [
    "Home",
    "Matharpada Rd",
    "Amit Naik Marg",
    "Caesar Cross Rd No. 1",
    "Caesar Rd (Alternative)",
    "Swami Vivekanand Rd",
    "Model Co-op Bank",
    "Andheri Subway Rd",
    "Dr. S. Radha Krishnan Rd",
    "MVLU College"
]
```

Distances according to the attached route information

```
route1_distance = 2.2
```

```
route2_distance = 2.3
```

DISPLAY ROUTES

```
print("ROUTE 1")
```

```
print(" -> ".join(route1))
```

```
print("Distance:", route1_distance, "km")
```

T101

RAJDEEPP M PARAB

TY.BSc.CS

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

```
print("ROUTE 2")
print(" -> ".join(route2))
print("Distance:", route2_distance, "km")

# COMPARE ROUTES
print("ROUTE COMPARISON")
print("Route 1 Distance:", route1_distance, "km")
print("Route 2 Distance:", route2_distance, "km")
if route1_distance < route2_distance:
    print("Route 1 is shorter by",
          round(route2_distance - route1_distance, 2), "km.")
elif route2_distance < route1_distance:
    print("Route 2 is shorter by",
          round(route1_distance - route2_distance, 2), "km.")
else:
    print("Both routes have equal distance.")
```

```
# BFS
start = "Home"
goal = "MVLU_College"
result = ""
def BFS(city, cityq, visitedq):
    global result
    if city == start:
        result = result + " " + city
    for eachcity in dict_gn[city].keys():
        if eachcity == goal:
            result = result + " " + eachcity
            return
        if eachcity not in cityq.queue and \
            eachcity not in visitedq.queue:
            cityq.put(eachcity)
            result = result + " " + eachcity
    visitedq.put(city)
    if not cityq.empty():
        BFS(cityq.get(), cityq, visitedq)
```

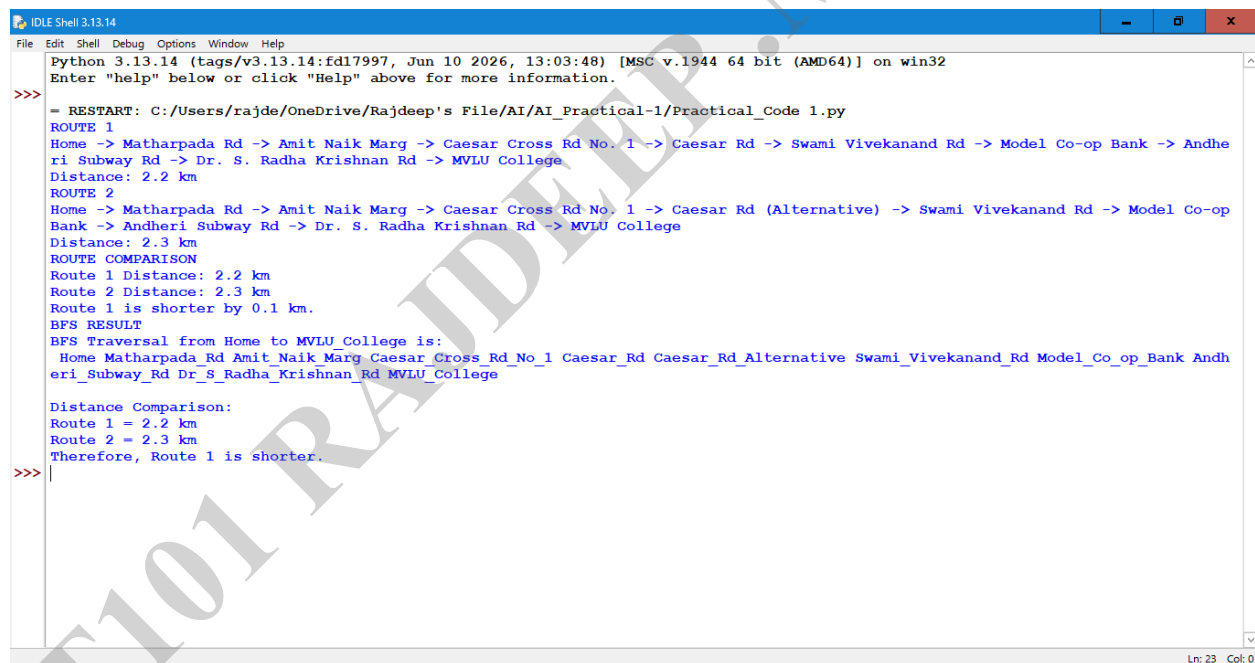
```
# MAIN
def main():
    cityq = Q.Queue()
    visitedq = Q.Queue()
    BFS(start, cityq, visitedq)
```

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: T101 ARTIFICIAL INTELLIGENCE Practical

```
print("BFS RESULT")
print("BFS Traversal from", start, "to", goal, "is:")
print(result)
print("\nDistance Comparison:")
if route1_distance < route2_distance:
    print("Route 1 =", route1_distance, "km")
    print("Route 2 =", route2_distance, "km")
    print("Therefore, Route 1 is shorter.")
elif route2_distance < route1_distance:
    print("Route 1 =", route1_distance, "km")
    print("Route 2 =", route2_distance, "km")
    print("Therefore, Route 2 is shorter.")
else:
    print("Both routes have equal distance.")
main()
```

Output :-



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/rajde/OneDrive/Rajdeep's File/AI/AI_Practical-1/Practical_Code 1.py
ROUTE 1
Home -> Matharpada Rd -> Amit Naik Marg -> Caesar Cross Rd No. 1 -> Caesar Rd -> Swami Vivekanand Rd -> Model Co-op Bank -> Andheri Subway Rd -> Dr. S. Radha Krishnan Rd -> MVLU College
Distance: 2.2 km
ROUTE 2
Home -> Matharpada Rd -> Amit Naik Marg -> Caesar Cross Rd No. 1 -> Caesar Rd (Alternative) -> Swami Vivekanand Rd -> Model Co-op Bank -> Andheri Subway Rd -> Dr. S. Radha Krishnan Rd -> MVLU College
Distance: 2.3 km
ROUTE COMPARISON
Route 1 Distance: 2.2 km
Route 2 Distance: 2.3 km
Route 1 is shorter by 0.1 km.
BFS RESULT
BFS Traversal from Home to MVLU College is:
Home Matharpada Rd Amit Naik Marg Caesar Cross Rd No. 1 Caesar Rd Caesar Rd Alternative Swami Vivekanand Rd Model Co-op Bank Andheri Subway Rd Dr S Radha Krishnan Rd MVLU College
Distance Comparison:
Route 1 = 2.2 km
Route 2 = 2.3 km
Therefore, Route 1 is shorter.
>>>
```